

Def. Let  $G$  be a group, and  $H \leq G$  be a subgroup.  
 Given an element  $g \in G$ , define left-Coset of  $H$  as  

$$gH \stackrel{\text{def}}{=} \{gh \mid h \in H\}.$$

Lemma. Let  $G$  be a group, and  $H \leq G$  be a subgroup. Then, for each  $g_1, g_2 \in G$ ,  
 $g_1H = g_2H$  if and only if  $g_1^{-1}g_2 \in H$ .

Proof. Given  $g_1, g_2 \in G$ , suppose that  $g_1H = g_2H$ . Then, since  $g_2 \in g_2H = g_1H$  ( $1 \in H$ ),  
 $g_2 = g_1h$  for some  $h \in H$ , thus  $g_1^{-1}g_2 \in H$ .

Conversely, let  $g_1, g_2 \in G$  be given st  $g_1^{-1}g_2 \in H$ . That is,  $g_1h = g_2$  for some  $h \in H$ .  
 Given  $g_1h_1 \in g_1H$ ,  $g_1h_1 = g_2h^{-1}h_1 \in g_2H$ . Given  $g_2h_2 \in g_2H$ ,  $g_2h_2 = g_1hh_2 \in g_1H$ .  $\square$

Q. In left coset sense,  $g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H$  hold?

A. No! Ss,  $H = \langle (1, 2) \rangle$ ,  $g_1 = (1\ 3)$  and  $g_2 = (1\ 2\ 3)$

Thm. Lagrange Thm

If  $H$  is a subgroup of a group  $G$ , then  $|G|/|H| = [G:H]$

Proof. Given a subgroup  $H \leq G$ , define a relation on  $G$  as;

$$x \sim y \Leftrightarrow xH = yH$$

Then, it is an equivalence relation, so a Collection  $\{gH \mid g \in G\}$  forms a partition.  
 Moreover, for each  $g \in G$ , a map

$$h \mapsto gh \quad (h \mapsto gh)$$

is onto by definition and one to one by the cancellation law, so  $|H| = |gH|$ .